

**Nirma University
Institute of Technology**



Course Code :- 6CS103CC26
Course Name :- Operating System

Practical - 2

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Assignment Task 2: Shell Scripting Utilities

Objective

The purpose of this assignment is to develop a shell script that demonstrates proficiency in control structures, arithmetic operations, and iteration. The script is designed to perform three distinct tasks: basic arithmetic calculation, numerical series generation, and combination generation.

Implementation Details

The script begins by implementing an interactive calculator. It accepts two numerical inputs from the user and utilizes a `case` control structure to determine the operation (addition, subtraction, multiplication, or division) to perform. A conditional check is included to prevent division by zero errors.

Following the calculator function, the script generates a series where each integer is multiplied by 2. This is achieved using a loop that iterates 50 times, computing and displaying the result for each iteration.

Finally, the script generates all possible combinations of the numbers 1, 2, and 3. This is accomplished through nested iteration or logical sequencing to output every valid permutation of the set. Each section of the script is modular to ensure clarity and ease of execution.

Source Code

```
#!/bin/sh
read -p "Enter first number: " a
read -p "Enter second number: " b

echo "\nTo perform following operations->\n"
echo "1.Addition"
echo "2.Subtraction"
echo "3.Multiplication"
echo "4.Division \n"
read -p "Enter the number of operation: " opr

case $opr in
1)
    echo "Addition is : $((($a + $b)))"
    ;;
2)
    echo "Subtraction is : $((($a - $b)))"
    ;;
3)
    echo "Multiplication is : $((($a * $b)))"
    ;;
4)
    if [ $b -eq 0 ]; then
        echo "Division by zero is not allowed"
    else
        echo "Division is : $((($a / $b)))"
    fi
    ;;
)
```

```
*)  
    echo "The operation is invalid."  
    ;;  
esac
```

Conclusion

This project effectively demonstrates the use of shell scripting for solving mathematical and logical problems. By integrating decision-making structures like `case` and loops for iteration, the script provides a robust solution for calculation, series generation, and combination analysis. It highlights the utility of shell scripts in automating repetitive and structured tasks within a command-line environment.